# Solar Radiation Analysis and Prediction

## Summary

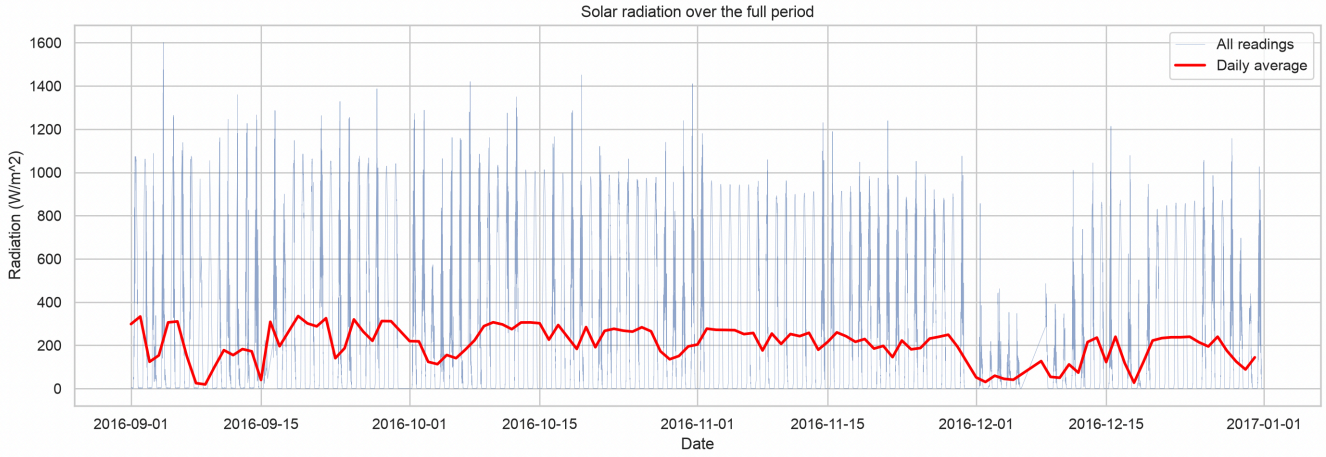
Solar radiation peaks around solar noon (12:00, mean 727.5 W/m²) and tracks temperature most closely (r = 0.73).

The Voyage-123 log holds 32,686 readings taken every ~5 minutes from 1 Sep to 31 Dec 2016 (118 days, local time). It has no missing values or duplicate rows. Temperature is in °F, pressure in inHg, humidity in %, wind speed in mph and radiation in W/m².

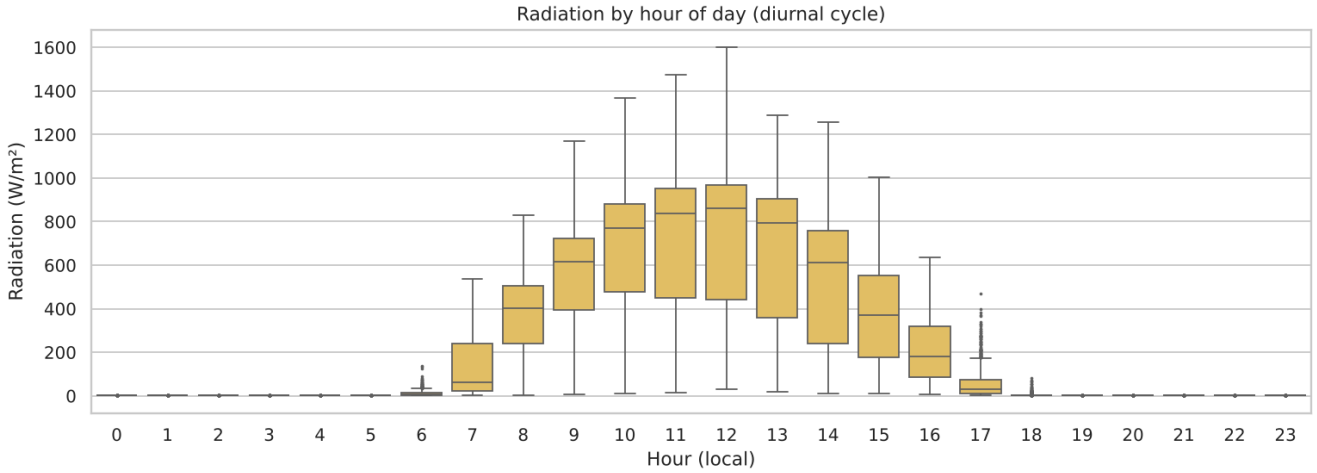
| Question | Answer |
|---|---|
| Time of highest radiation | 11:00–13:00, peak hour 12:00 |
| Strongest related parameter | Temperature, r = 0.73 |
| Date with highest radiation | 4 Sep 2016, 1601.26 W/m² at 12:15:04 |
| Peak after sunrise | 6 h 07 min (sunrise 06:08) |

## Part A — Data visualisation

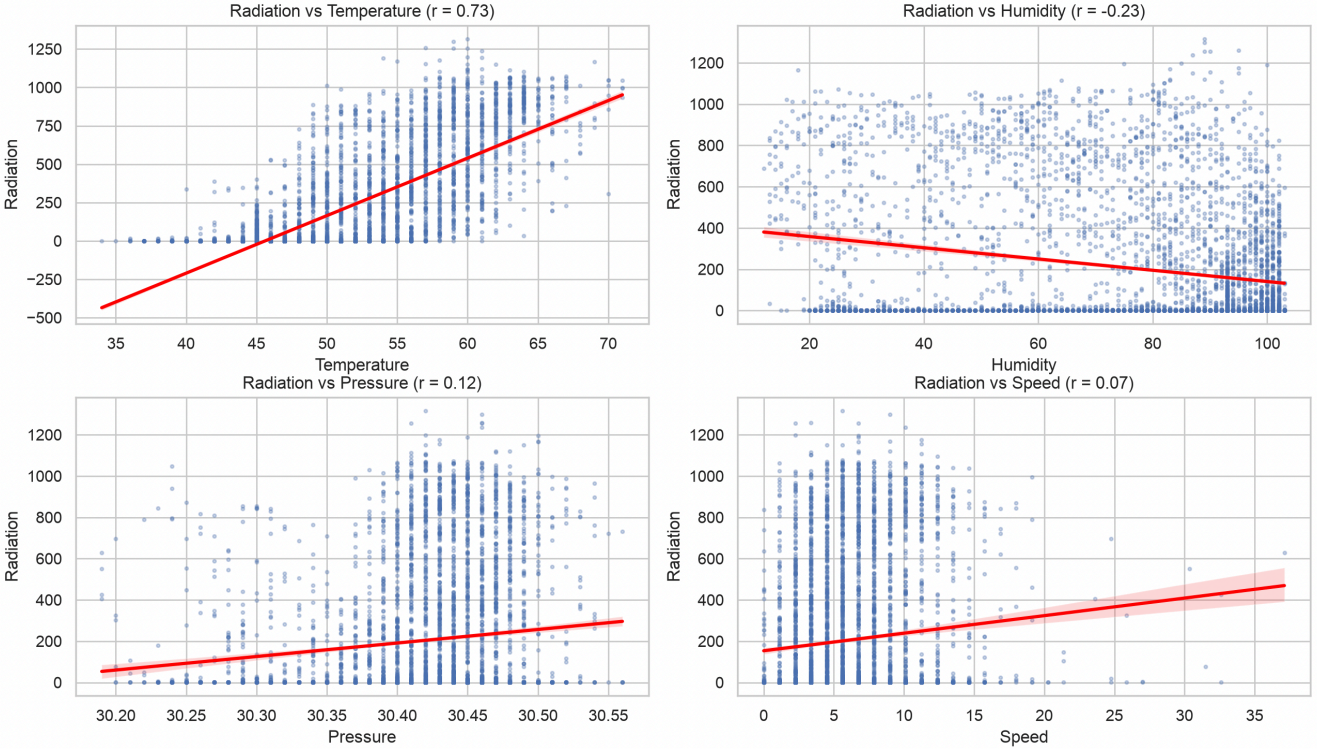
**A1. Radiation over time.** Every day follows the same rise-and-fall cycle. The daily average (red) holds around 200–330 W/m² from September to November, dips during cloudy spells such as 7–9 Sep, and falls below 100 W/m² in early December.



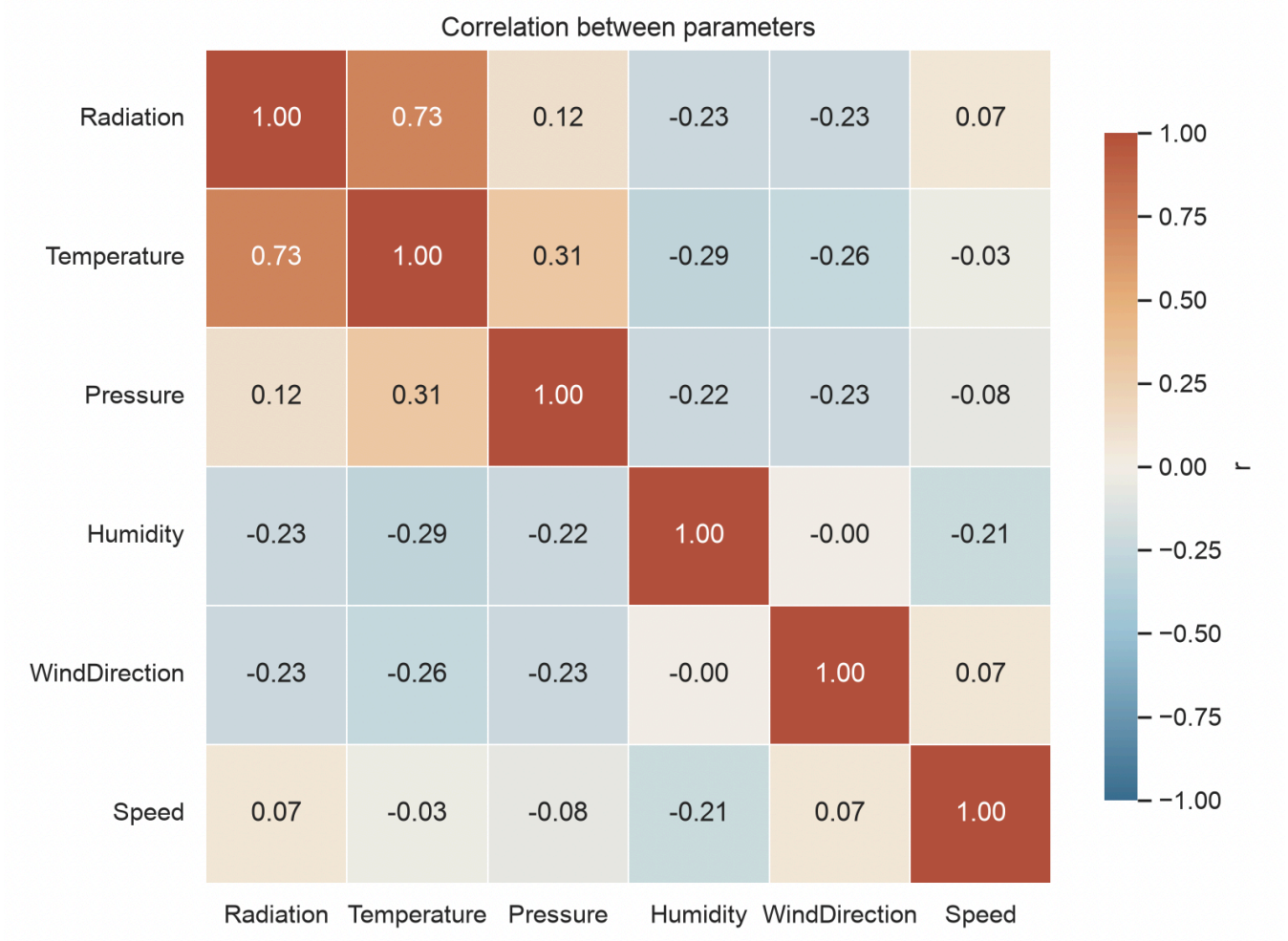

**A2. Radiation by hour of day.** The boxplot shows the diurnal cycle: near zero from 18:00 to 06:00, steepest rise between 07:00 and 09:00, widest spread around noon where clouds cause the most variation.



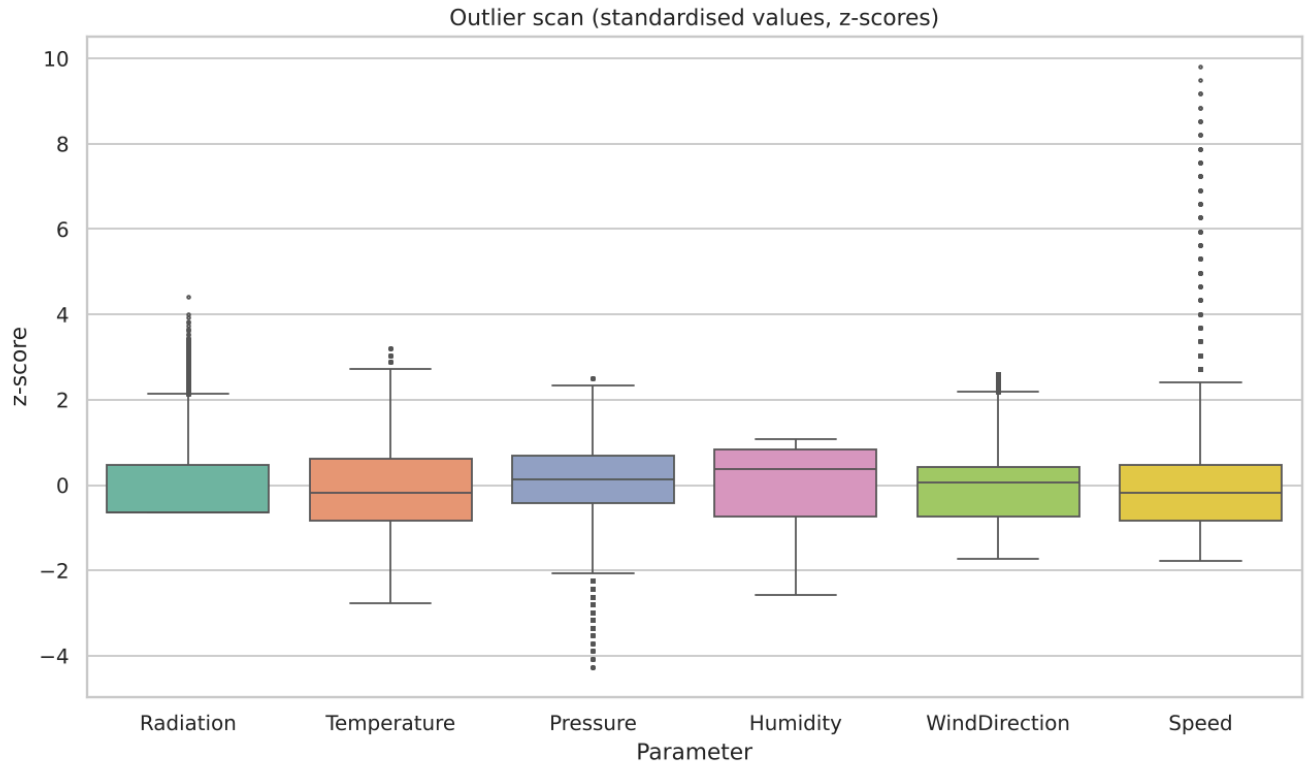

**A3. Radiation against temperature, humidity, pressure and wind speed.** The red line is a linear fit with its confidence band. Temperature has the steepest and clearest upward trend; humidity slopes downward; pressure and wind speed are close to flat, and the wind-speed slope above 20 mph rests on only a few readings.




**A4. Correlation matrix.**




**A5. Outlier scan.** All parameters standardised to z-scores so their spread can be compared on one axis. Wind speed has the longest upper tail; radiation is right-skewed because half the readings are at night.




# Part B — Drawing inferences

**When is radiation highest?** Between 11:00 and 13:00. Mean radiation is 713.7 W/m² at 11:00, 727.5 W/m² at 12:00 and 649.5 W/m² at 13:00. From 19:00 to 05:00 it sits at 1.3–1.4 W/m², the sensor's night-time floor. The daily peak falls a median 5.76 h after sunrise, which matches solar noon (A2, C1).

**Which parameter is most strongly related?** Temperature, with r = 0.73. The other correlations are weak: wind direction −0.23, humidity −0.23, pressure 0.12, wind speed 0.07 (A4). Using daylight readings only, so that the day/night effect is removed, temperature still gives r = 0.65 and humidity strengthens to −0.39.

**Radiation, temperature and humidity together.** Radiation heats the ground and air, so temperature rises with it: mean radiation climbs from about 0 W/m² below 45 °F to about 1,000 W/m² at 70 °F (A3). Humidity moves the other way. Warmer air holds more moisture, so relative humidity falls in the late morning (lowest at 10:00) and rises again at night, often above 95 %. Readings with high humidity and low radiation are mostly overcast or rainy periods.

**Outliers and what explains them.**

- **Radiation above the solar constant.** 7 readings exceed 1,361 W/m², the most sunlight that reaches the top of the atmosphere; the highest is 1,601.26 W/m² on 4 Sep at 12:15. All 7 occur near noon with humidity of 69–98 %. The likely cause is cloud enhancement: sunlight reflected off cloud edges adds to the direct beam for a few minutes.

- **Humidity above 100 %.** 4,311 readings show 101–103 %, which is physically impossible; this is sensor calibration drift during saturated, foggy nights.
- **Wind speed spikes.** 149 readings exceed 20 mph and the maximum is 40.5 mph, against a mean of 6.24 mph. These are short gusts, probably from passing storms.
- **Non-zero night radiation.** Night readings average 1.41 W/m² with a maximum of 15.75 W/m², which reflects sensor offset and twilight rather than real sunlight.
- **Data gaps.** Six days have fewer than 200 readings out of a normal 288, including 9 Sep with 123 and 8 Dec with 154. These gaps lower the daily averages for those days.

**Bonus insight: the season ends with a step, not a slide.** Daylight-hours mean radiation is 468 W/m² in September, 468 in October and 480 in November, then falls 36 % to 306 W/m² in December. Day length shrinks only gradually (12.24 h to 10.95 h), so the sudden December drop is driven mostly by more frequent cloudy days, rather than by the sun's lower angle alone. For the base, this means power planning should allow for a sharp December shortfall.

## Part C — Solar day analysis

The highest radiation was recorded on 4 Sep 2016: 1,601.26 W/m² at 12:15:04, 6 h 07 min after sunrise.

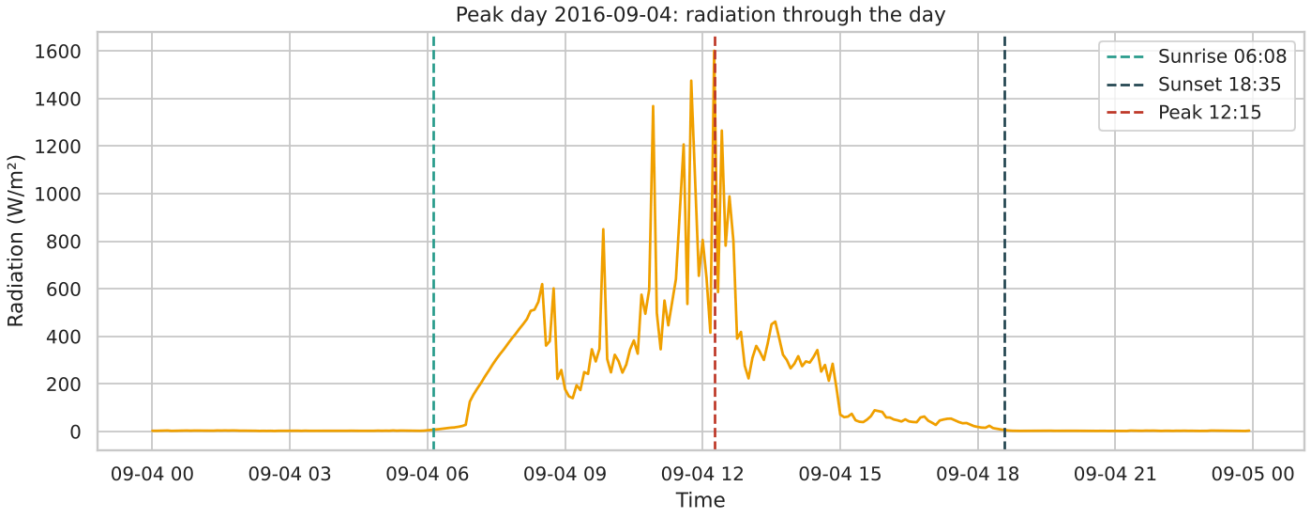
**Datetime conversion.** The `Data` column carries the date with a dummy `12:00:00 AM`, and `Time` carries the clock time. Both are local time (`UNIXTime` is UTC), so the datetime is built as date part of `Data` + `Time` with `pd.to_datetime`. The full daily report (max, average, reading count, sunrise, sunset for all 118 days) is saved as `daily_solar_report.csv`.

**Peak day details**

| Measure | Value |
|---|---|
| Date | 4 Sep 2016 |
| Max radiation | 1,601.26 W/m² |
| Time of max | 12:15:04 |
| Temperature | 61 °F |
| Humidity | 93 % |
| Wind speed | 9.0 mph |
| Sunrise · sunset | 06:08 · 18:35 |

| Measure | Value |
|---|---|
| Peak after sunrise | 6 h 07 min 04 s (6.12 h) |
| Peak before sunset | 6 h 19 min 56 s |

Solar noon that day was about 12:21, so the peak landed 6 minutes before it. Across all days the peak comes a median 5.76 h after sunrise (middle half: 5.40–6.25 h). The 93 % humidity at the peak supports the cloud-enhancement explanation in Part B: the day's average radiation was only 153.58 W/m², a cloudy day with brief bright spikes.




### Top 5 days by average radiation

| Rank | Date | Avg radiation (W/m²) | Max radiation (W/m²) |
|---|---|---|---|
| 1 | 19 Sep 2016 | 335.50 | 1,085.75 |
| 2 | 2 Sep 2016 | 333.47 | 1,062.01 |
| 3 | 22 Sep 2016 | 325.52 | 1,053.17 |
| 4 | 25 Sep 2016 | 319.92 | 1,075.29 |
| 5 | 28 Sep 2016 | 312.40 | 1,029.30 |

All five are in September, when days are longest, and all have full coverage (275–284 readings), so the ranking is not distorted by data gaps. The day with the single highest reading does not appear: a high maximum does not mean a high average.